

NPTEL Course on
"Computer System Optimization for ML"

Lecture #17

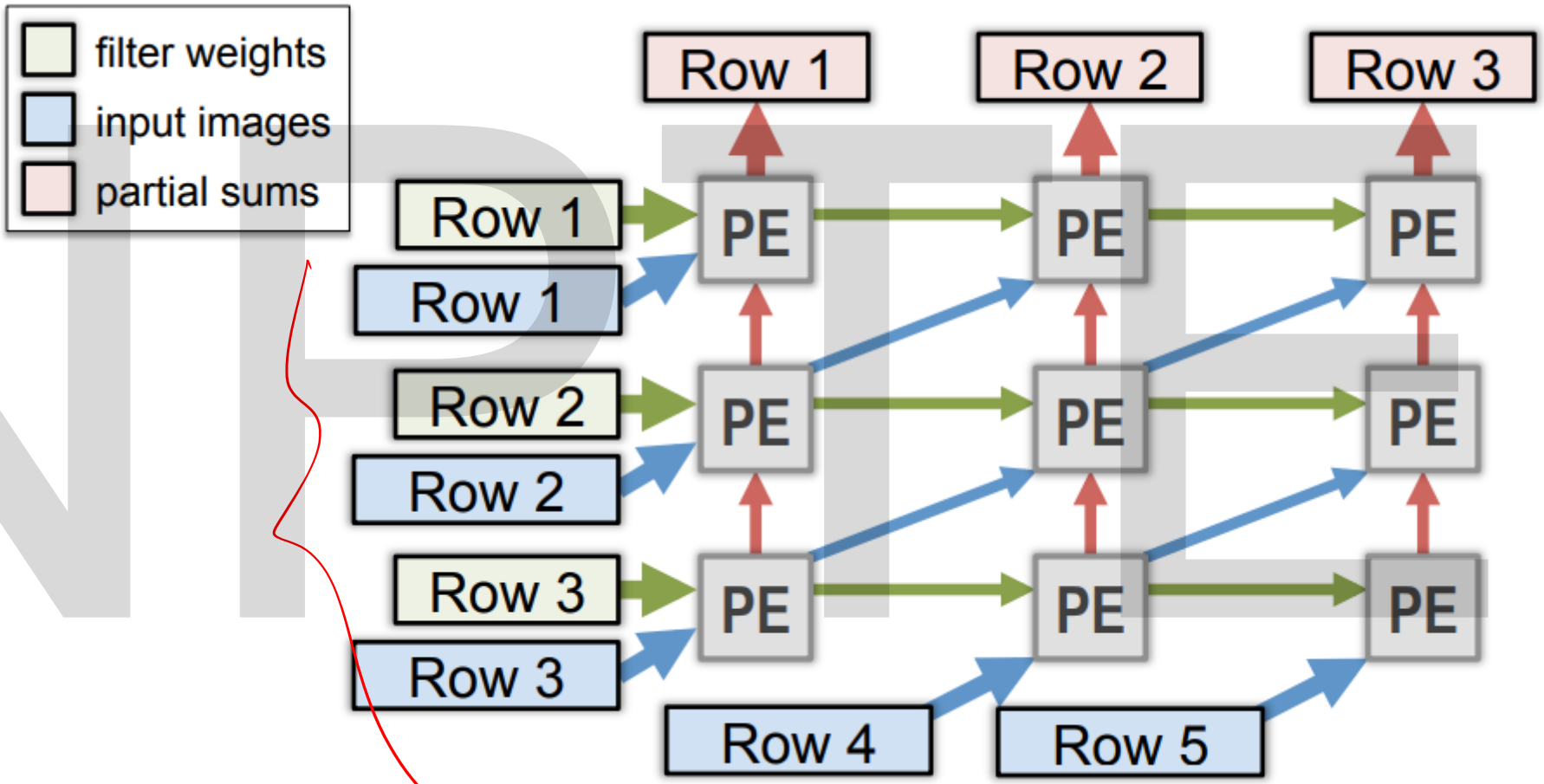
Agenda

- **Recap**
- **Eyeriss-V2 [2]**
- **DNN Pruning**

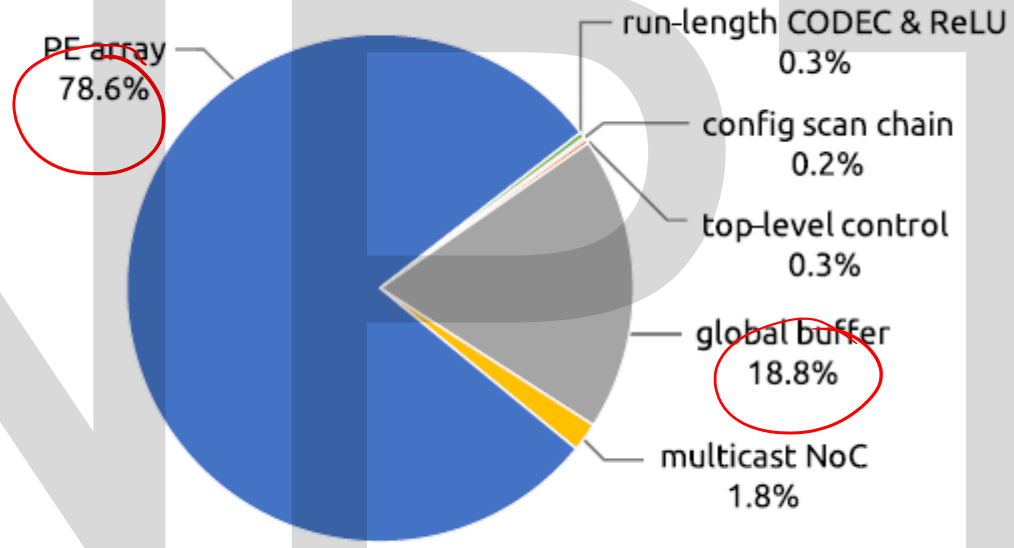
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[2] Chen, Yu-Hsin, et al. "Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices." IEEE Journal on Emerging and Selected Topics in Circuits and Systems 9.2 (2019): 292-308.

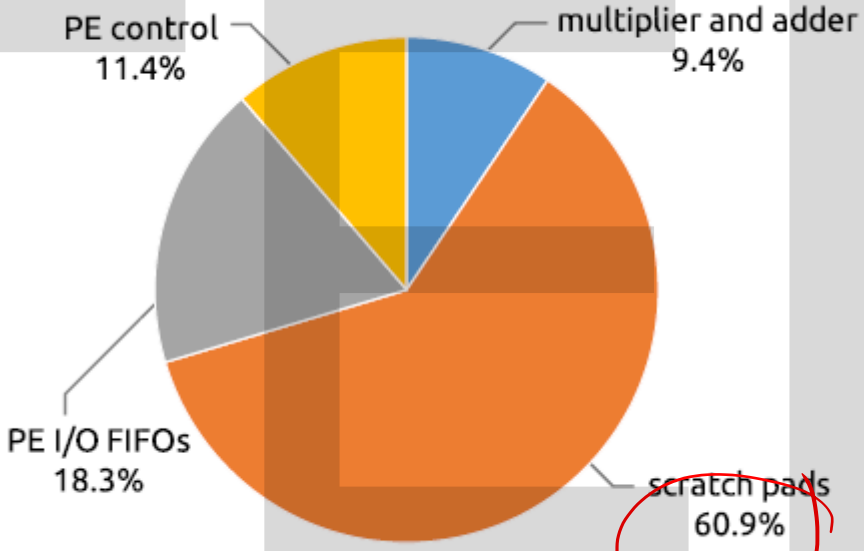
Row Stationary Dataflow



Results: Area Breakdown of Eyeriss



(a) **Eyeriss core**



(b) **PE**

NDA

Agenda

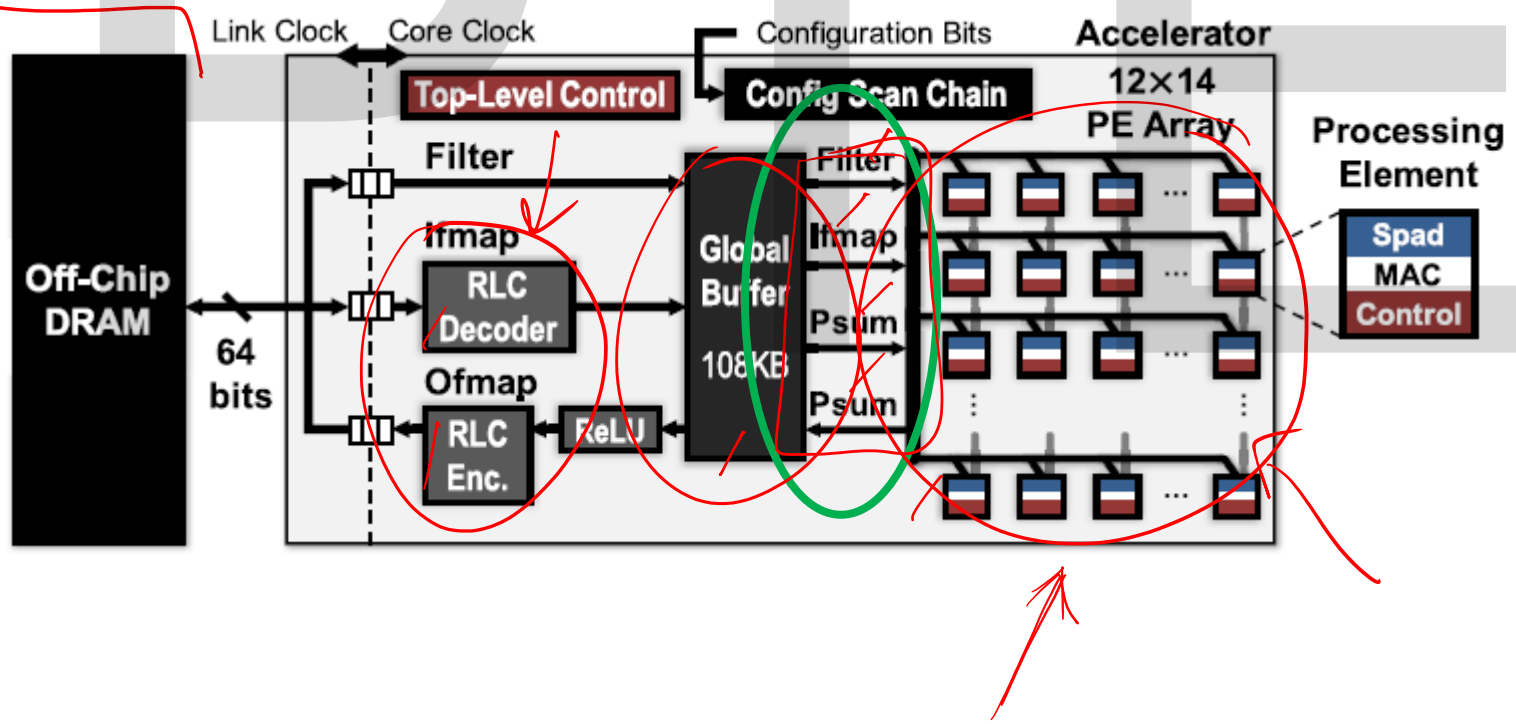
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Eyeriss-V2

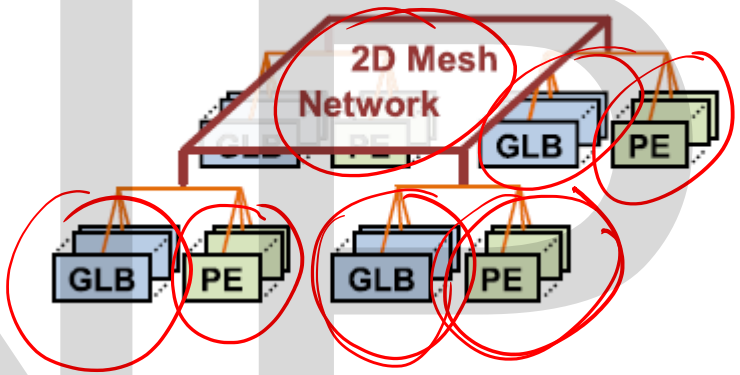
- **Eyeriss-V2: A Flexible Accelerator for Emerging DNNs on Mobile Devices**
- **What is wrong with Eyeriss?**
  - Think from communication perspective
  - Too many data transfer between global buffer and PE array

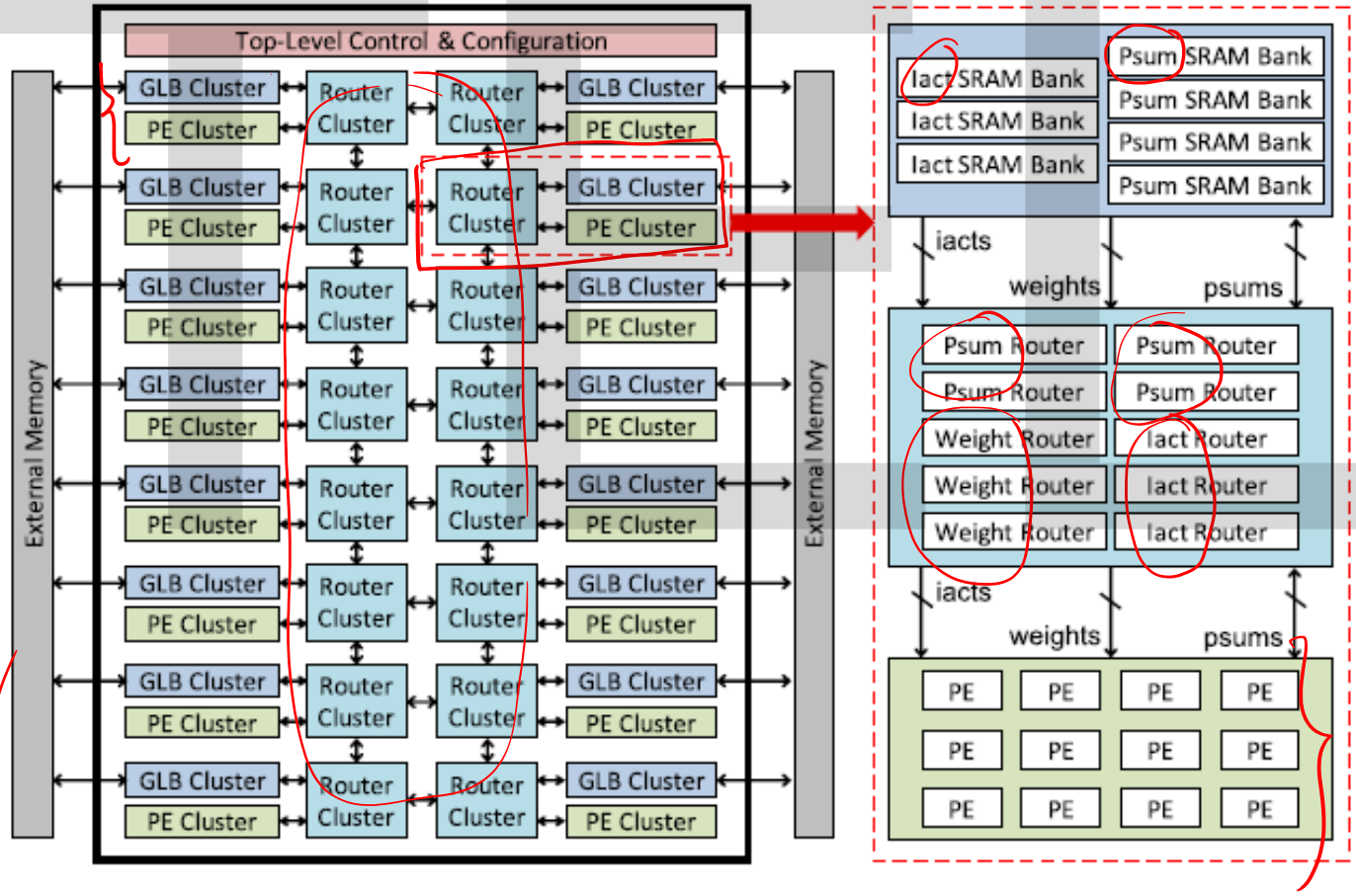


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Any Idea?

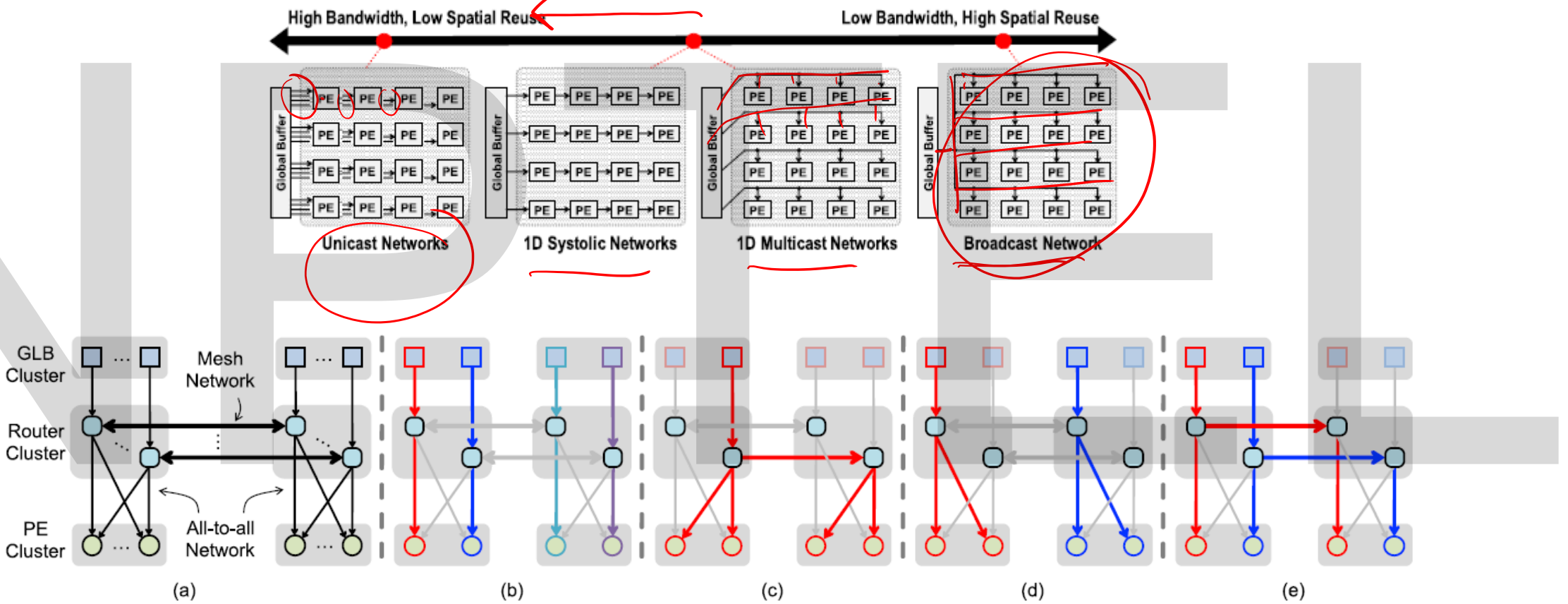
- **Distribute the communication**





Advantage

Common NoC Designs (until the paper was published)



Eyeriss-V2: a) High-level structure of the hierarchical mesh network (HM-NoC), and its different operating modes: (b) High bandwidth mode, (c) High reuse mode, (d) grouped-multicast mode, and (e) interleaved-multicast mode. In each mode, the colored arrows show the routing path; different colors denote the path for unique data.

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Topics in DNN Pruning

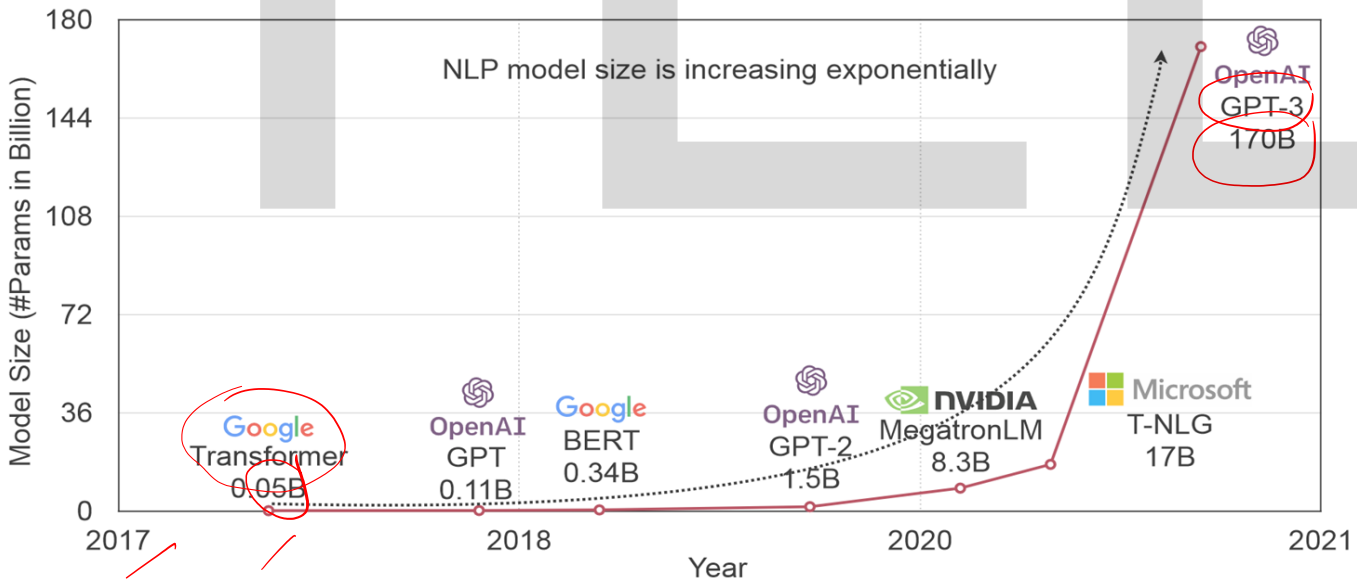
- The need for smaller models
- How to select which parameter to prune
- Develop ‘a’ recipe for DNN pruning
- Problems due to pruning
- Role of underlying hardware on pruning

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Slide Courtesy: Sourav Mazumdar, IISc

The Need for Smaller Models

- **Increasing size of neural networks**
  - Overparameterization
- **High cost of deployment and sustained inference**
- **Unable to run on edge devices**
  - High inference time
  - High memory
  - Energy consumption



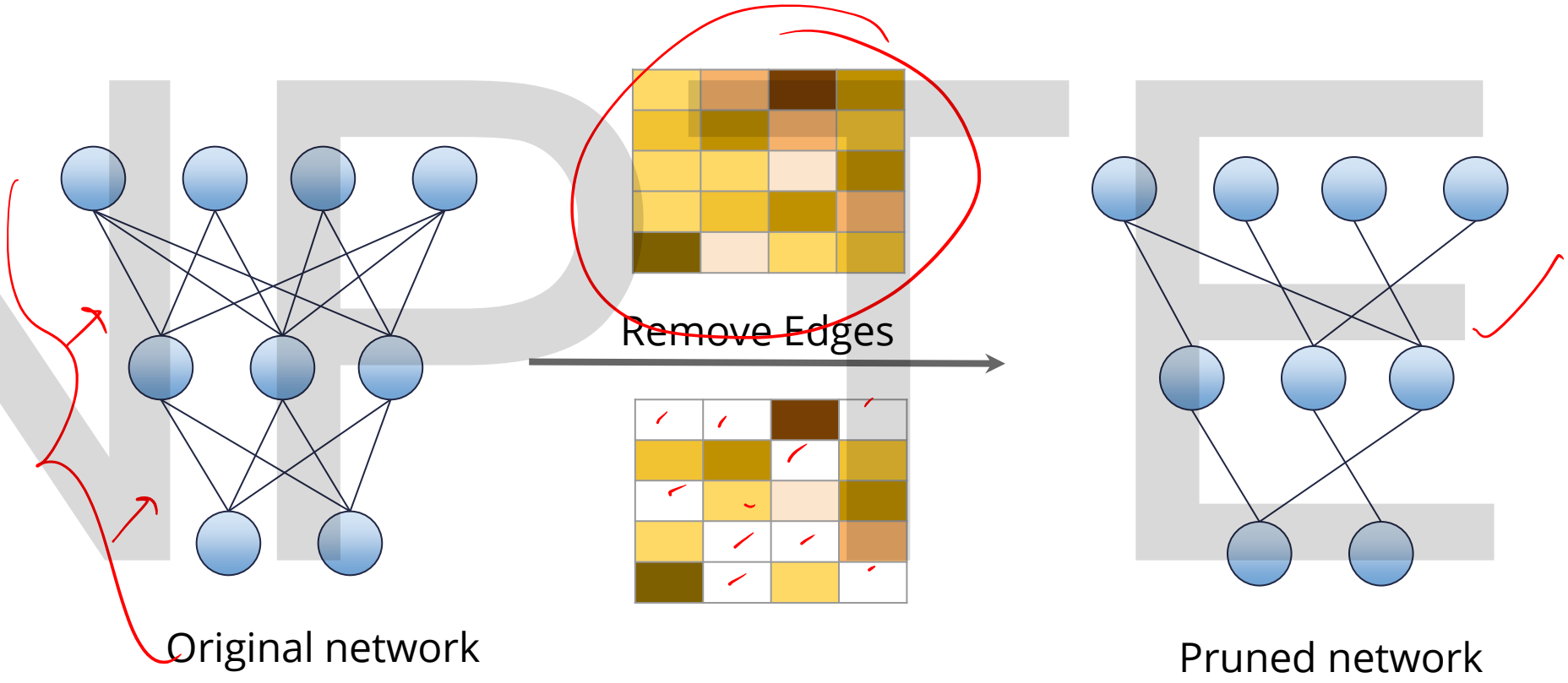
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In search of Smaller Models

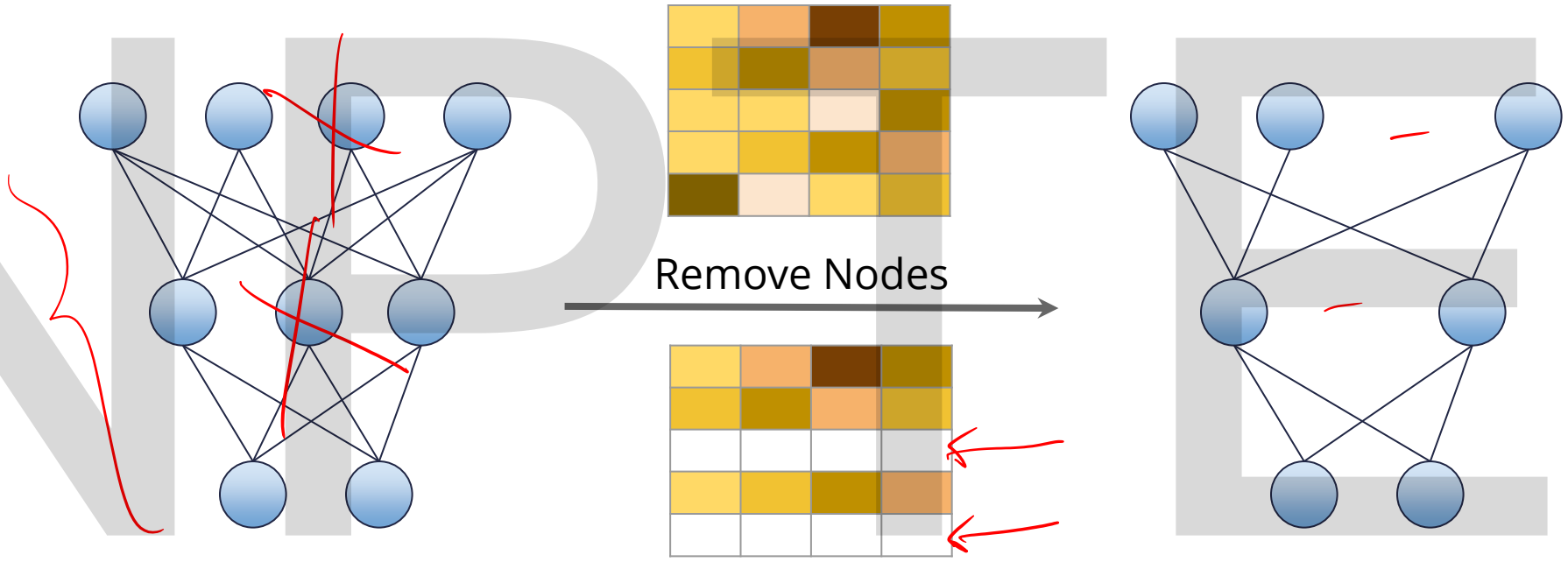
- **'somehow' create compressed models with faster execution times and without losing accuracy much**
- **Knowledge Distillation (KD)**
  - A smaller (student) model is made to learn from a bigger (teacher) model
- **KD suffers from the drawback of having to define a completely new model and train that model from scratch**
- **Better approach**
  - Compress existing models without training any new model from scratch

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Overview of DNN Pruning (1)



Overview of DNN Pruning (2)

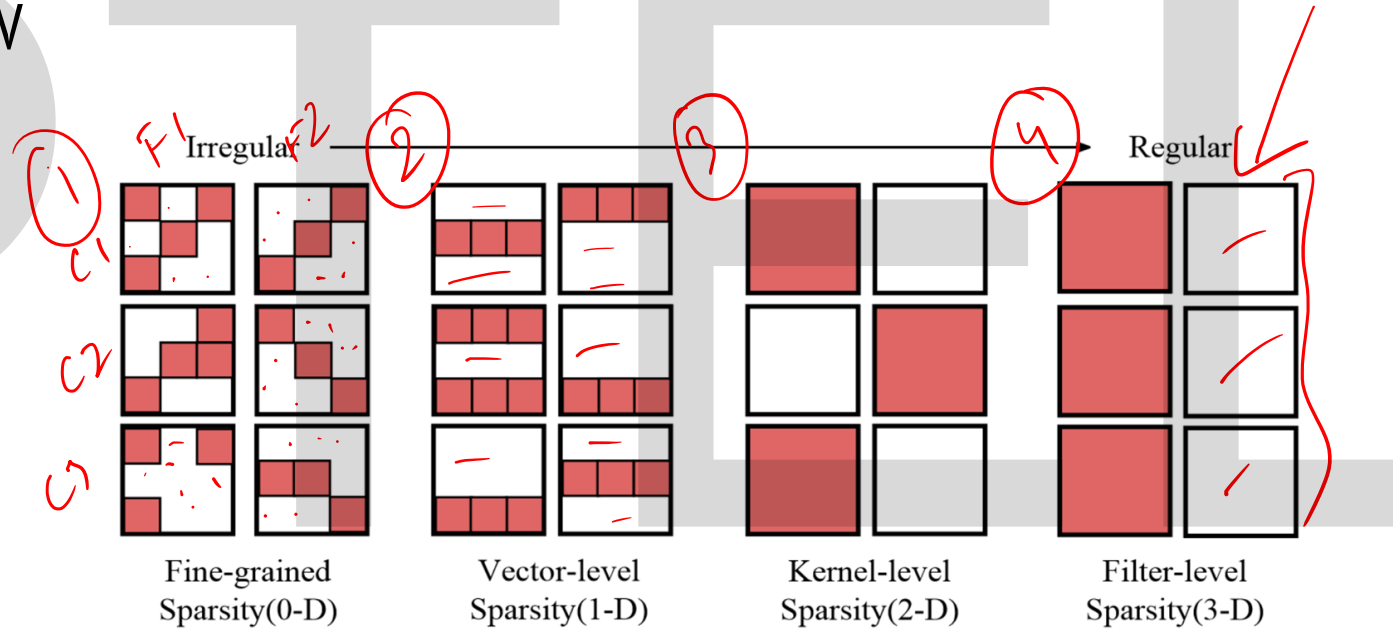


Original network

Pruned network

Pruning Granularity in DNNs (1)

- **Fine Grained: Individual Weights**
- **Vector Level**
  - Sub-kernel vectors of size W
- **Kernel Level**
  - Kernels of shape H x W
- **Filter Level**
  - Filter of shape C x H x W

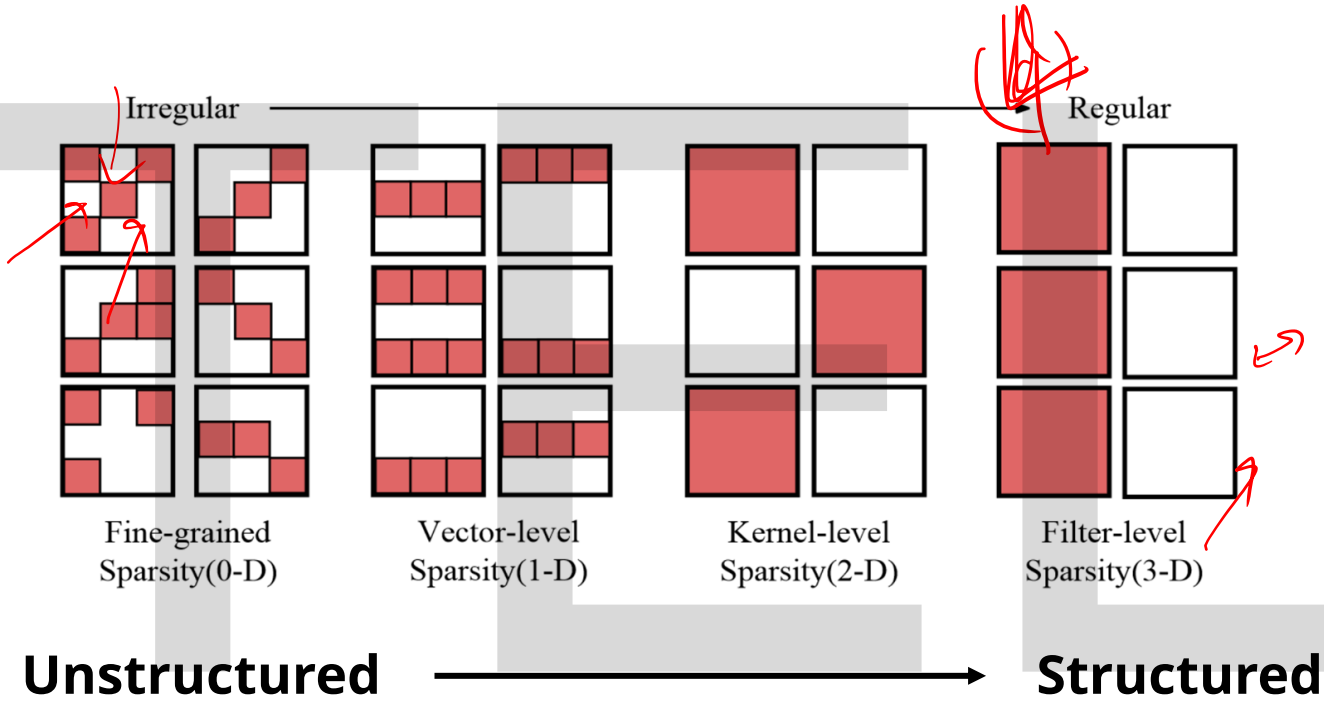


Unstructured → **Structured**

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Pruning Granularity in DNNs (2)

- **Unstructured**
  - Finer control over selecting elements to prune
  - Can provide better sparsity to accuracy ratio, i.e., smaller network with higher accuracy
  - Difficult to implement
- **Structured**
  - Less control over selecting elements to prune
  - Relatively less sparsity to accuracy ratio
  - Easier to implement



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Mathematical Definition of DNN Pruning

- DNN pruning can be defined as

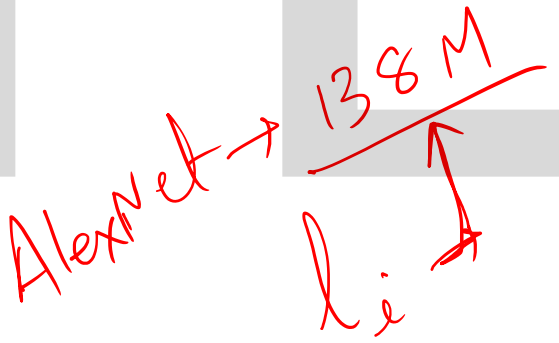
$$argmin_p L(W_p, D) s.t. \phi(W_p) \leq C$$

- L is the loss/objective function of the DNN
- D is the target dataset
- W is the original weights, $W_p \in W$ is a subset of weights remaining after pruning
- $\phi(.)$ is some function which maps the network to some constraint C such as inference time, FLOPS, memory etc

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Which Parameters to Prune?

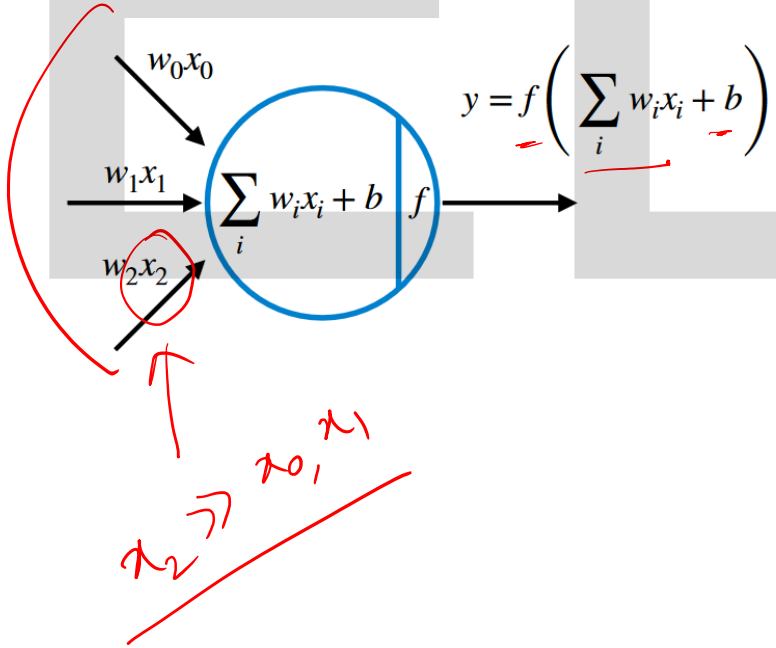
- **Select the less important parameters to prune**
  - Define a saliency term, which can map each parameter to some importance score
  - The importance score can be the loss in performance (degradation of accuracy) when removing that particular parameter
  - Remove the lowest salient parameters from the network



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Magnitude-based Pruning

- f(.) = ReLU(.)
- W = [10, -8, 0.1]
- $Y = f(10x_0 - 8x_1 + 0.1x_2)$
- If you can afford to remove one weight parameter, which one do you remove?
- What is the basis for your selection?



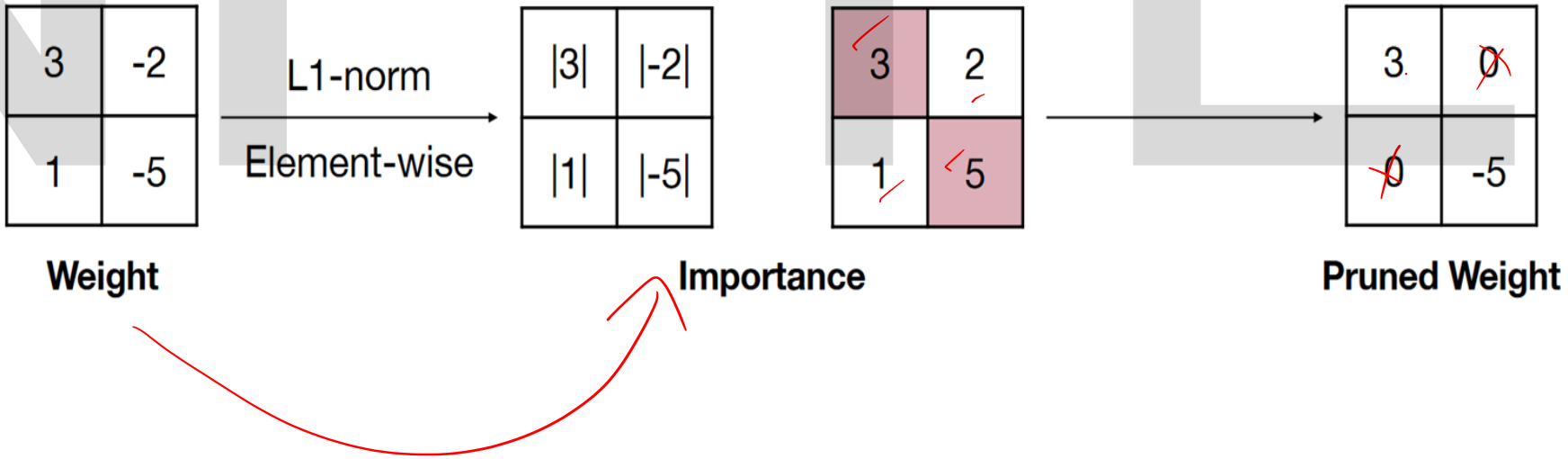
Importance $\propto$ |W|

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Magnitude-based Pruning

- Remove the parameters whose absolute values are closer to 0
- For element-wise pruning

Importance = |W|

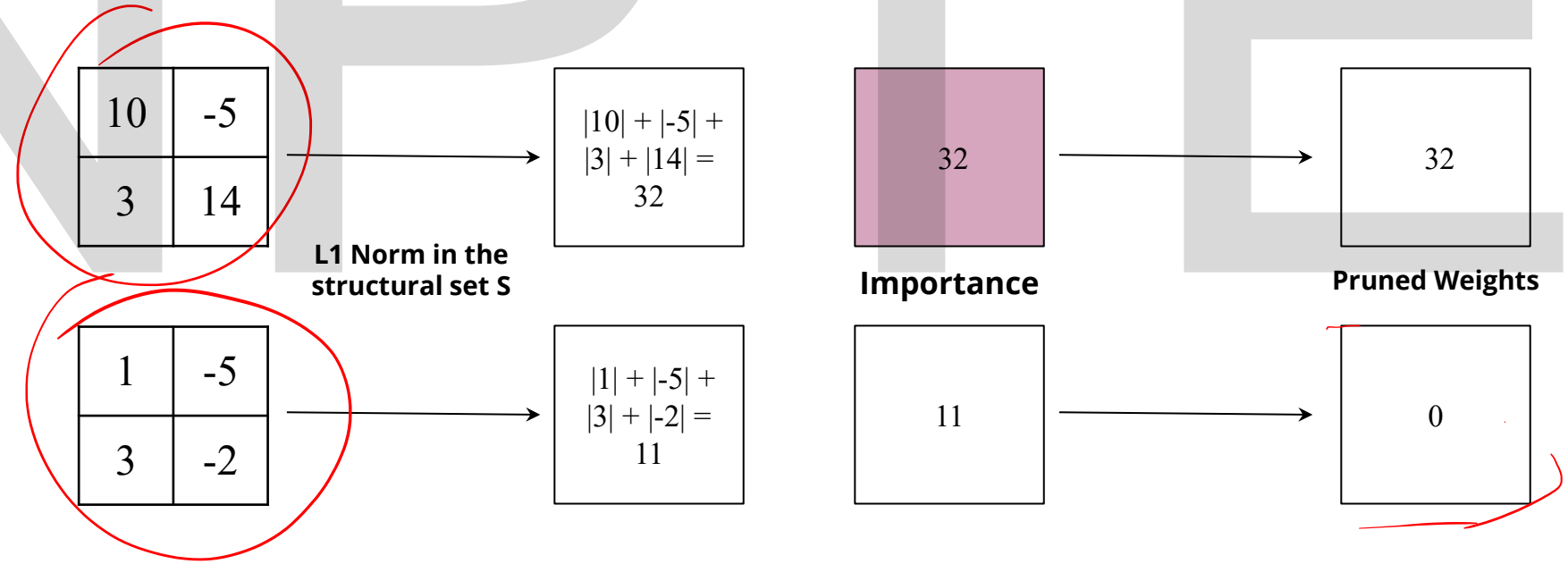


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Magnitude-based Pruning

- Remove the parameters whose absolute values are closer to 0
- For element-wise pruning

$$\text{Importance} = \sum_{i \in S} |W_i|$$



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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)